RIG-SMOOTH QUOTIENT ALGEBRAS ARE STABLE UNDER SUFFICIENTLY CLOSE DEFORMATIONS OF THEIR PRESENTATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a three-constant bound n, a lifting of the mod-I^n comparison to a genuine homomorphism, and an isomorphism proof combining surjectivity by completeness with injectivity via the associated graded and Krull intersection, all inside the delivered span

Editorial note. The label lemma-close-enough names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Before the final part of the argument the author writes 'we suggest the reader find their own proof of this' and then immediately provides the complete proof (surjectivity, associated graded, Krull intersection); per the quality criterion this reading suggestion is not an omission, and the verification is delivered verbatim.

Allowed standard prerequisites (editorial).

- The rig-smooth homomorphism definition via annihilation of Ext^1 of the completed naive cotangent complex
- The approximation lemmas for complexes over adic rings from More on Algebra, giving exactness of the perturbed presentation and an isomorphism of associated graded rings for n large
- The uniformity integers d and q of the associated graded, and the lifting lemma producing a homomorphism congruent to a prescribed one modulo a prescribed power

Source reference. Original lemma 0315: sources/stacks-chow/tags.txt, source lines 3972–3972. Prerequisite pointer invoked at proof line(s) 1434; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00IP: sources/stacks-chow/tags.txt, source lines 687–687. Prerequisite pointer invoked at proof line(s) 1447; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06RN: sources/stacks-chow/tags.txt, source lines 8647–8647. Prerequisite pointer invoked at proof line(s) 1439; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0GAI: sources/stacks-restricted/pinned-original-source.tex, source lines 831–831. Prerequisite pointer invoked at proof line(s) 1383; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAQ: sources/stacks-restricted/pinned-original-source.tex, source lines 1167–1167. Prerequisite pointer invoked at proof line(s) 1419; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0GA6: sources/stacks-chow/tags.txt, source lines 20326–20326. Prerequisite pointer invoked at proof line(s) 1401; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07VE: sources/stacks-chow/tags.txt, source lines 10004–10004. Prerequisite pointer invoked at proof line(s) 1388; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07VF: sources/stacks-chow/tags.txt, source lines 10005–10005. Prerequisite pointer invoked at proof line(s) 1389; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let I be an ideal of a Noetherian ring A. Let $r \geq 0$ and write $P = A[x_1, \ldots, x_r]$ the I-adic completion. Consider a resolution*

$$P^{\oplus t} \xrightarrow{K} P^{\oplus m} \xrightarrow{g_1, \ldots, g_m} P \to B \to 0$$

of a quotient of P. Assume B is rig-smooth over (A, I). Then there exists an integer n such that for any complex

$$P^{\oplus t} \xrightarrow{K'} P^{\oplus m} \xrightarrow{g'_1, \ldots, g'_m} P$$

with $g_i - g'_i \in I^n P$ and $K - K' \in I^n Mat(m \times t, P)$ there exists an isomorphism $B \to B'$ of A-algebras where $B' = P/(g'_1, \ldots, g'_m)$.

Proof. (A) By Definition 0GAI we can choose a $c \geq 0$ such that I^c annihilates $\mathrm{Ext}^1_B(NL^\wedge_{B/A}, N)$ for all B-modules N.

(B) By More on Algebra, Lemmas 07VE and 07VF there exists a constant $c_1 = c(g_1, \ldots, g_m, K)$ such that for $n \geq c_1 + 1$ the complex

$$P^{\oplus t} \xrightarrow{K'} P^{\oplus m} \xrightarrow{g'_1, \ldots, g'_m} P \to B' \to 0$$

is exact and $\mathrm{Gr}_I(B) \cong \mathrm{Gr}_I(B')$.

(C) Let $d_0 = d(\mathrm{Gr}_I(B))$ and $q_0 = q(\mathrm{Gr}_I(B))$ be the integers found in Local Cohomology, Section 0GA6.

We claim that $n = \max(c_1 + 1, q_0 + (d_0 + 1)c, 2(d_0 + 1)c + 1)$ works where c is as in (A), c_1 is as in (B), and q_0, d_0 are as in (C).

Let $g'_1, \ldots, g'_m$ and K' be as in the lemma. Since $g_i = g'_i \in I^n P$ we obtain a canonical A-algebra homomorphism

$$\psi_n : B \longrightarrow B'/I^n B'$$

which induces an isomorphism $B/I^nB \to B'/I^nB'$. Since $\mathrm{Gr}_I(B) \cong \mathrm{Gr}_I(B')$ we have $d_0 = d(\mathrm{Gr}_I(B'))$ and $q_0 = q(\mathrm{Gr}_I(B'))$ and since $n \geq \max(q_0 + (1+d_0)c, 2(d_0+1)c+1)$ we may apply Lemma 0GAQ to find an A-algebra homomorphism

$$\varphi : B \longrightarrow B'$$

such that $\varphi \bmod I^{n-(d_0+1)c}B' = \psi_n \bmod I^{n-(d_0+1)c}B'$. Since $n-(d_0+1)c > 0$ we see that φ is an A-algebra homomorphism which modulo I induces the isomorphism $B/IB \to B'/IB'$ we found above. The rest of the proof shows that these facts force φ to be an isomorphism; we suggest the reader find their own proof of this.

Namely, it follows that φ is surjective for example by applying Algebra, Lemma 0315 part (1) using the fact that B and B' are complete. Thus φ induces a surjection $\mathrm{Gr}_I(B) \to \mathrm{Gr}_I(B')$ which has to be an isomorphism because the source and target are isomorphic Noetherian rings, see Algebra, Lemma 06RN (of course you can show φ induces the isomorphism we found above but that would need a tiny argument). Thus φ induces injective maps $I^eB/I^{e+1}B \to I^eB'/I^{e+1}B'$ for all $e \geq 0$. This implies φ is injective since for any $b \in B$ there exists an $e \geq 0$ such that $b \in I^eB$, $b \notin I^{e+1}B$ by Krull's intersection theorem (Algebra, Lemma 00IP). This finishes the proof. $\square$